②

tysolo

$u(t) = 50 \sin(314t)$ [V]

$i(t) = 0{,}1 \sin(314t + \frac{\pi}{4})$ [A]

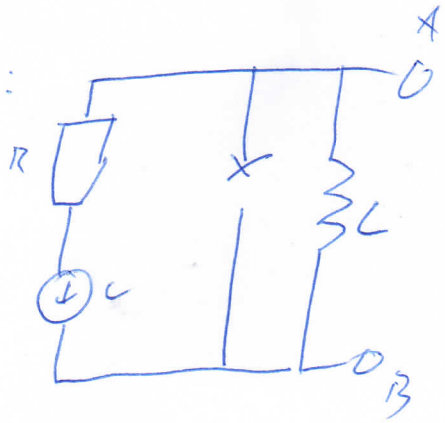
$\hat{U} = 50 e^{j \cdot 0} = 50\,V$

$\hat{I} = 0{,}1 e^{j\frac{\pi}{4}}\,A$

$$R_i = \frac{R\, j\omega L}{R + j\omega L} = \frac{1000 \cdot j \cdot 314 \cdot 1{,}591}{1000 + j \cdot 314 \cdot 1{,}591} = 200 + 400j$$

$\hat{U}_i = \hat{U}_i' + \hat{U}_i''$

$\hat{U}_i'$:



$\rightarrow U_i' = U \cdot \frac{R}{R_i}$

$\hat{U}_i'' = I \cdot R_i$

$\hat{U}_i = \hat{U} - \hat{U} \cdot \frac{R}{R_i} + I R_i$